

Percussion Welding | Advantages and Disadvantages

Percussion welding is a resistance welding process where in coalescence is produced simultaneously over the entire area of abutting surfaces by heat obtained from an arc produced by a rapid discharge of stored electrical energy. This is a self-timing spot welding method. The parts to be welded are clamped in copper jaws of the welding machine in which one clamp is fixed and other one is movable. The movable clamp is backed up against the pressure from a heavy spring. The jaws act as electrodes. Heavy electric current is connected to the workpieces. Now, the movable clamp is released rapidly and it moves forward at high velocity. When the two parts are approximately 1.6 mm apart, a sudden discharge of electrical energy is released thereby causing an intense arc between two surfaces. The arc is extinguished by the percussion blow of the two parts coming together with sufficient force to complete in 0.1 second. No upset or flash occurs at the weld. This method is primarily employed to join dissimilar metals. This method is limited to small areas of about 150 to 300 mm².

Welding energy, $E = 1/2 CV^2$

where

E = energy in watt-seconds (Joules)

C = capacitance in farads

V = voltage.

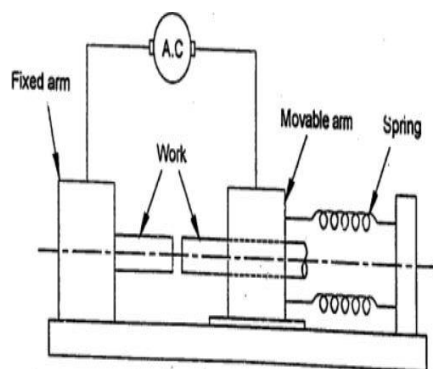


Figure 2.17 Percussion welding

Amount of energy required to make joint depends on the following factors.

- (i) Cross-sectional area of joint
- (ii) Properties of work metal or metals
- (iii) Depth to which metal is melted on workpieces.

Advantages of Percussion Welding:

1. The time cycle involved is very short.
2. Shortness of arc limits melting and heating.
3. Heat-treated and cold worked materials can be welded without annealing.
4. No filler metal is required.
5. No cast structure is produced at interface.
6. Charging rate is low and controlled.
7. It can tolerate some contamination on faying surface.
8. Welding of dissimilar metal and copper to steel is possible.
9. These welders have a long wear life for welding jaws due to very short relatively low current welding pulse.
10. Welding of metals with high melting point such as tungsten, molybdenum etc. are possible by this method.

Limitations of Percussion Welding:

1. The welding process is limited to butt joints.
2. Total area is limited.
3. Similar metals can usually be joined more economically by other processes.
4. The process is usually confined for joining of dissimilar metals not normally considered weldable.
5. Welding is typically dirtier and less smooth than resistance welding.
6. With nib start percussive arc welding, a starting nib must be cut onto workpieces.
7. The workpieces must be free of oil or dirt.

Applications:

Percussion welding process can be used for welding a large number of dissimilar metals. It is used for very specialized applications and the process is entirely automatic. Metals which can be percussion welded include, copper alloys, aluminium

alloys, nickel alloys, low-carbon steel, medium-carbon steel and stainless steels. This process is used for welding satellite tips to tools, copper to aluminium or stainless steel, silver contact tips to copper, cast iron to steel, lead-in-wires on electric lamps and zinc to steel. Gold, silver, copper-tungsten, silver-tungsten and silver-cadmium oxide percussion welded to copper alloys for commonly used assemblies for electric contacts.

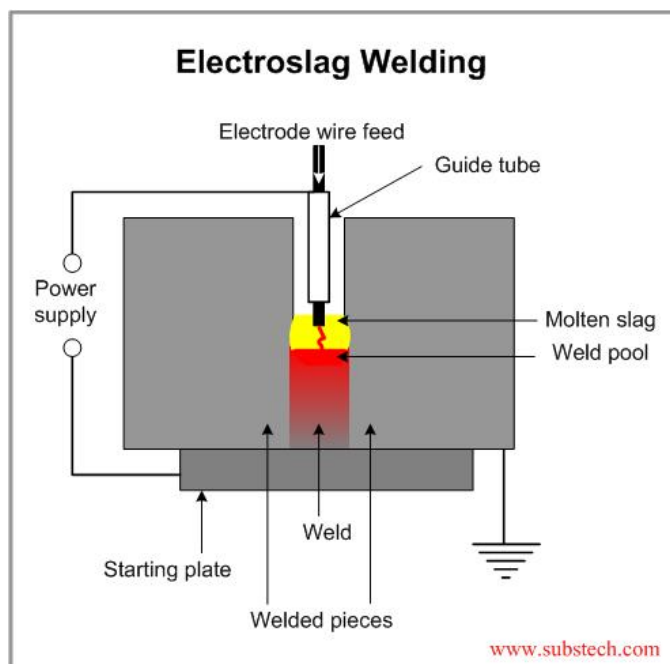
Electroslag Welding Process – Principle, Main Parts, Working, Advantages and Disadvantages with Application

Electroslag Welding Process is a process in which the job is welded through the molten slag covering which is a result of a high quantity of heat which is generated by the circulation of electric current through the electrode and the job.

- Robert K Hopkins discovered this welding process in February 1940 at Paton Institute, Kiev, USSR.
- This welding process is a single pass high productive welding process that is used for welding thick (25-300 mm) materials in vertical or close to vertical positions.

Principle:

Electroslag welding process works on the principle of heat generation due to arc and electric resistance. Arc is made between the welding electrode and the workpiece which starts melting the filler metal to fill the filler cavity. Now the heat is produced due to electric resistance when the current passes through the surface. The heat further starts melting the filler metal which is continuously fed from the roller. This filler wire melts and fills the weld and makes it a strong joint.



1. **Electrodes:** Generally two types of electrodes that are solid and metal-cored are used. Though solid electrodes are more popular than the metal-cored electrodes.
2. **Flux:** Flux is the most important consumable material of Electroslog Welding. In its molten state, it transforms the electrical energy into heat energy which helps in melting the electrode wire and the base metal to form a weld joint. It is also required to protect the molten weld metal from the atmosphere and to ensure stability. The flux in its molten state is required to conduct electricity but at the same time, it should offer sufficient resistance to its flow for generating enough heat to do welding.

Arcing results when the resistance is less than the minimum required quantity. The slag must also have minimum viscosity so that it does not be too thick to stop good circulation and nor too thin to result in excessive leakage.

3. Electrode Guide Tube: It is used to guide the electrode wire at the desired position where the welding is to be done.

Working of Electroslog Welding

It works as follows:

- Firstly current flows from welding electrode to base plate. This establishes an arc between electrode and base plate which heats the flux. This heat which is produced during arc formation results in melting the filler metal and deposits into the weld cavity.
- Now the cooled copper shoe starts its function of solidifying the filler metal into the weld cavity. This is done to avoid the flowing of weld metal outside.
- As the filler metal solidifies into the weld cavity, the current flows through it. Then it generates heat due to electric resistance. This heat is further used to continue the melting of the filler metal into the weld cavity. Which means that heat is regenerated, which results in less waste of heat or energy.
- Roller arrangement continuously provides the filler metal.
- During welding of the metals both copper shoe and feed mechanism moves upward until the whole cavity is formed.

- This will create a strong joint in a single pass. The single or multi-pass weld is used according to plate thickness.

Advantages:

1. The cooling rate is very low so there is no problem of cold cracking.
2. There is no problem of slag inclusion or porosity.
3. The process is semi-automatic and faster.
4. Heavier section can be welded in a single pass.
5. High productivity can be achieved.
6. Low cost for joint preparation.

Disadvantages:

1. Too high heat input to base.
2. High temperature of welding needs cooling arrangement.
3. Slow rate of cooling give columnar grain in weld.
4. When the heat input is very high then the weld quality can be poor, including low toughness caused by the coarse grains in the fusion zone and the heat-affected zone.
5. In Electroslag welding, it has some tendency towards hot cracking and notch sensitivity in the heat affected zone.
6. It is restricted to vertical position welding, because of large molten metal pools and slag.
7. It is difficult to close cylindrical welds
8. It tends to produce large grain sizes..
9. If you are creating joints below 60 mm than **Submerged Arc Welding (SAW)** is more economical than electroslag welding process.

Application:

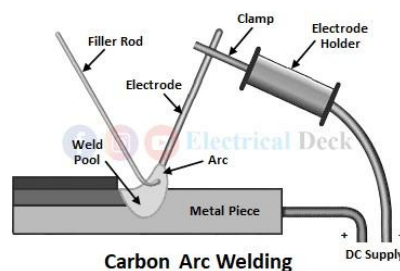
1. It is used in heavy industries where plate thickness is up to 80 mm to be joined.
2. Welding of thick walled large diameter pipes is done by this welding process.
3. Welding of storage tanks is done by it.
4. It is used to construct big and thick parts of ships.

Carbon Arc Welding - Working, Advantages & Applications

Carbon arc welding is the first electric welding process invented by Russian inventor Nikolay Benardos in 1881. In this welding method, an electrode made up of carbon or graphite is used to initiate an electric arc. The welding is done by heating up the metal pieces with the electric arc struck between the carbon electrode and the workpiece to be welded.

Carbon Arc Welding :

The below shows the arrangement of the carbon arc welding process. It consists of an electrode holder to which a carbon electrode is attached. The arc is maintained between the metal piece and the carbon electrode held in the holder. The arc is struck by touching the electrode with the metal piece momentarily and then taking away it at a definite distance apart.



The arc is allowed to impinge on the surface to be welded till a molten pool forms and then the holder is steadily moved along the joint. Filler metal and flux may or may not be used depending upon the type of joint and material to be welded.

In carbon arc welding, the electrode is usually connected to the negative terminal and the workpiece to be welded is made as positive. So that less heat will be generated at the electrode tip than at the metal piece because there is a tendency of fusing carbon particles from the electrode and mix in the weld which causes brittleness.

Thus dc supply is preferred for carbon arc welding since there is no fixed polarity in the case of ac supply. Generally, a voltage of about 30 Volts is used in case of in case ac supply and 40 Volts in case of dc supply.

Electrodes Used in Carbon Arc Welding :

In carbon arc welding electrodes made of carbon or graphite with a diameter of 2.5mm and a length of 300mm are used. Graphite electrode has more brittleness, hardness, and longer life compared to carbon electrode. They can withstand higher current densities but it is difficult to control the arc column.

Due to longer life and low resistance, graphite electrode is preferred over carbon electrode. The arc produced between the two electrodes heats the metal to melting temperature which is about 3200°C on the positive electrode.

The carbon or graphite electrodes used are generally non-consumable electrodes, thus there is a requirement for additional filler material if in case needed for welding which depends upon the type of joint and materials to be welded. Thus this type of welding is normally employed where there is no requirement for filler material. Even though the electrode is of a non-consumable type, vapourization and oxidization cause the electrode to disintegrate gradually.

Advantages of Carbon Arc Welding :

The advantages of carbon arc welding are,

- Heat input to the metal piece can be easily controlled by changing the arc length.
- This type of welding is suitable for joining thinner pieces.
- Total welding cost is less compared to other welding processes.
- It is quite a clean welding method with a simple process and good welding skills can be acquired in a short time.
- Both ferrous and non-ferrous metals can be welded.
- The carbon arc welding process can be done automatically using electronic controllers.
- For soldering, brazing and braze welding, etc, carbon arc welding is the best method of heat source.

Disadvantages of Carbon Arc Welding :

The disadvantages of carbon arc welding are,

- There are chances of carbon being transferred from electrode to weld thus causing a harder weld deposit in the case of ferrous materials.
- In the absence of proper electrode geometry, arc blow results which give poor welds with blow holes.
- Separate filler metal is needed which when used slows down the welding speed.
- The current required to attain welding temperature is almost double the current required in metal arc welding.
- Carbon arc welding can be efficiently and economically done with dc supply.

Applications of Carbon Arc Welding :

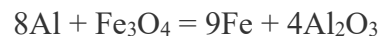
- For welding steel, aluminum, nickel, copper, and a good number of other alloys.
- Carbon arc can also be employed for brazing, preheating, and post-heating of welded joints.
- Welding of stainless steel of thinner gauges.
- Welding of Monel metal using suitable coated filler material.

Thermite welding

Thermit Welding is a **welding process** utilizing heat generated by exothermic chemical reaction between the components of the thermit (a mixture of a metal oxide and **aluminum** powder). The molten metal, produced by the reaction, acts as a filler material joining the work pieces after **Solidification**.

Thermit Welding is mainly used for joining **steel** parts, therefore common thermit is composed from iron oxide (78%) and aluminum powder (22%).

The proportion 78-22 is determined by the chemical reaction of combustion of aluminum:



The combustion reaction products (iron and aluminum oxide) heat up to 4500°F (2500°C). Liquid iron fills the **sand (or ceramic) mold** built around the welded parts, the slag (aluminum oxide), floating up, is then removed from the weld surface.

Thermit Welding is used for repair of steel **casings** and **forgings**, for joining railroad rails, steel wires and steel pipes, for joining large cast and forged parts.

Advantages of Thermit Welding:

No external power source is required (heat of chemical reaction is utilized);

Very large heavy section parts may be joined.

Disadvantages of Resistance Welding:

Only ferrous (steel, chromium, nickel) parts may be welded;

Slow welding rate;

High temperature process may cause distortions and changes in **Grain structure** in the weld region.

Weld may contain gas (**Hydrogen**) and slag contaminations.

Solid State Welding (SSW)

Solid State Welding is a **welding process**, in which two work pieces are joined under a pressure providing an intimate contact between them and at a temperature essentially below the melting point of the parent material. Bonding of the materials is a result of **diffusion** of their interface atoms.

Advantages of Solid State Welding:

Weld (bonding) is free from microstructure defects (**pores**, non-metallic inclusions, **segregation** of **alloying** elements)

Mechanical properties of the weld are similar to those of the parent metals

No consumable materials (filler material, fluxes, shielding gases) are required

Dissimilar metals may be joined (**steel** - **aluminum alloy** steel - **copper alloy**).

Disadvantages of Solid State Welding:

Thorough surface preparation is required (degreasing, oxides removal, brushing/sanding)

Expensive equipment.

The following processes are related to Solid State welding:

Forge Welding (FOW)

Cold Welding (CW)

Friction Welding (FRW)

Explosive Welding (EXW)

Diffusion Welding (DFW)

Ultrasonic Welding (USW)

Forge Welding (FOW)

Forge Welding is a **Solid State Welding** process, in which **low carbon steel** parts are heated to about 1800°F (1000°C) and then **forged** (hammered).

Prior to Forge Welding, the parts are scarfed in order to prevent entrapment of oxides in the joint.

Forge Welding is used in general blacksmith shops and for manufacturing metal art pieces and welded tubes.

Advantages of Forge Welding:

- Good quality weld may be obtained;
- Parts of intricate shape may be welded;
- No filler material is required.

Disadvantages of Forge Welding:

- Only low carbon steel may be welded;
- High level of the operators skill is required;
- Slow welding process
- Weld may be contaminated by the coke used in heating furnace.

Cold Welding (CW)

Cold Welding is a Solid State Welding process, in which two work pieces are joined together at room temperature and under a pressure, causing a substantial deformation of the welded parts and providing an intimate contact between the welded surfaces.

As a result of the deformation, the oxide film covering the welded parts breaks up, and clean metal surfaces reveal. Intimate contact between these pure surfaces provide a strong and defectless bonding.

Aluminum alloys, Copper alloys, low carbon steels, Nickel alloys, and other ductile metals may be welded by Cold Welding.

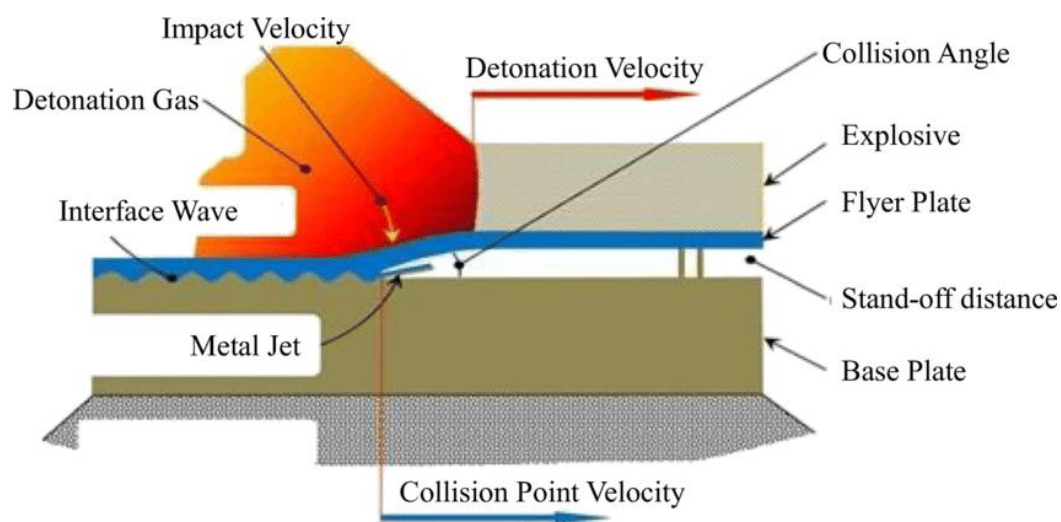
Cold Welding is widely used for manufacturing bi-metal steel - aluminum alloy strips, for cladding of aluminum alloy strips by other aluminum alloys or pure aluminum (Corrosion protection coatings). Bi-metal strips are produced by Rolling technology. Presses are also used for Cold Welding.

Cold Welding may be easily automated.

Explosive Welding (**shockwave**) (EXW)

Explosive Welding is a **Solid State Welding** process, in which welded parts (plates) are metallurgically bonded as a result of oblique impact pressure exerted on them by a controlled detonation of an explosive charge.

Explosion welding (EXW), also called explosive welding or shockwave welding, is a solid state welding process that involves accelerating one of the components at a high speed with the use of chemical explosives. The process uses the merging principle, which occurs in milliseconds and eliminates bulk heating or metal weakening.



One of the welded parts (base plate) is rested on an anvil, the second part (flyer plate) is located above the base plate with an angled or constant interface clearance.

Explosive charge is placed on the flyer plate. Detonation starts at an edge of the plate and propagates at high velocity along the plate.

The maximum detonation velocity is about 120% of the material sonic velocity.

The slags (oxides, nitrides and other contaminants) are expelled by the jet created just ahead of the bonding front.

Most of the commercial metals and alloys may be bonded (welded) by Explosive Welding.

Dissimilar metals may be joined by Explosive Welding:

Copper to steel;

Nickel to steel;

Aluminum to steel;

Tungsten to steel;

Titanium to steel;

Copper to aluminum.

Advantages of Explosive Welding

Large surfaces may be welded;

High quality bonding: high strength, no distortions, no porosity, no change of the metal microstructure;

Low cost and simple process;

Surface preparation is not required.

Disadvantages of Explosive Welding:

Brittle materials (low ductility and low impact toughness) cannot be processed;

Only simple shape parts may be bonded: plates, cylinders;

Thickness of flyer plate is limited - less than 2.5" (63 mm);

Safety and security aspects of storage and using explosives.

Explosive Welding is used for manufacturing clad tubes and pipes, pressure vessels, aerospace structures, heat exchangers, bi-metal sliding bearings, ship structures, weld transitions, corrosion resistant chemical process tanks.

Diffusion Welding (DFW)

Diffusion Welding is a Solid State Welding process, in which pressure applied to two work pieces with carefully cleaned surfaces and at an elevated temperature below the melting point of the metals. Bonding of the materials is a result of mutual diffusion of their interface atoms.

In order to keep the bonded surfaces clean from oxides and other air contaminations, the process is often conducted in vacuum.

No appreciable deformation of the work pieces occurs in Diffusion Welding.

Diffusion Welding is often referred more commonly as Solid State Welding (SSW).

Diffusion Welding is able to bond dissimilar metals, which are difficult to weld by other welding processes:

Steel to tungsten;

Steel to niobium;

Stainless steel to titanium;

Gold to copper alloys.

Diffusion Welding is used in aerospace and rocketry industries, electronics, nuclear applications, manufacturing **composite materials**.

Advantages of Diffusion Welding:

Dissimilar materials may be welded (**Metals, Ceramics, Graphite**, glass);

Welds of high quality are obtained (no **pores**, inclusions, chemical **segregation**, distortions).

No limitation in the work pieces thickness.

Disadvantages of Diffusion Welding:

Time consuming process with low productivity;

Very thorough surface preparation is required prior to welding process;

The mating surfaces must be precisely fitted to each other;

Relatively high initial investments in equipment.

Classification of welding processes

Welding	
Fusion welding	Solid-state welding
1. Arc welding (SMAW, GMAW, TIG, SAW, FCAW, ESW, etc.) 2. Gas welding (AAW, OAW, OHW, PGW) 3. Resistance welding (RSW, RSEW, PW, PEW, FW, etc.) 4. Intense energy beam welding (PAW, EBW, LBW)	1. Cold Welding (CW) 2. Roll Welding (ROW) 3. Pressure Welding (PW) 4. Diffusion Welding (DFW) 5. Friction Welding (FRW) 6. Friction Stir Welding (FSW) 7. Forge Welding (FOW), etc.